

P8131 Spring 2026 Homework #8 Solution

Due on April 20 at 11:59pm

1. The data (HEALTH.xlsx) are from a randomized, controlled trial among women of childbearing age to evaluate the effects of an educational intervention. One response variable of interest is the participants self-rating of health status as either good or poor. The researchers would like to assess the effect of the intervention on self-rated health across the follow-up period, as well as whether these effects are influenced by the mothers age. There are $n = 80$ women enrolled in this trial. These data were measured at 4 points in time: randomization, 3 months, 6 months, and 12 months post-randomization. Each observation in the dataset contains values for the following variables (in this order):

ID: unique participant identification code

TIME: the visit number for this observation of this participant

1: corresponds to a participants visit at the time of randomization

2: corresponds to the 3 month post-randomization visit

3: corresponds to the 6 month post-randomization visit

4: corresponds to the 12 month post-randomization visit

TRT: the treatment group to which the participant has been randomized

Control

Intervention

HEALTH: participants self-rated level of health for this visit

Good

Poor

AGEGROUP: participants age group at time of randomization

15-24 (years old)

25 to 34 (years old)

35+ (years old)

- (a) Evaluate the bivariate, cross-sectional relationship between randomized group assignment and participants health self-rating at the time of randomization. Interpret and discuss these findings.
- (b) Perform a longitudinal data analysis across all study follow-up visits (but not at randomization) to describe the relationship of the participants self-ratings as a function of the effects of health self-rating at the baseline, treatment group, month post randomization, and age group as predictors. Fit a GEE with unstructured correlation structure. Interpret your results.
- (c) Fit a generalized linear mixed effects model with subject-specific random intercepts. Interpret your estimates. How are the interpretations different from the GEE model?

Solution:

- (a) We can propose a logistic regression model:

$$\log \frac{P(Y_i = 1)}{P(Y_i = 0)} = \beta_0 + \beta_1 I(Treatment)_i$$

Call:

```
glm(formula = HEALTH2 ~ TXT, family = binomial, data = health0
)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.157	-1.157	-1.028	1.198	1.335

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.04879	0.31244	-0.156	0.876
TXIntervention	-0.31412	0.45122	-0.696	0.486

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 110.10 on 79 degrees of freedom
Residual deviance: 109.62 on 78 degrees of freedom
AIC: 113.62

From the output, we know that treatment indicator is not significant, with P-value $0.486 > 0.05$. Hence we know there is no significant relationship between randomized group assignment and participants health self-rating at the time of randomization. This is as expected due to randomization.

- (b) The mean function of the GEE model is

$$\begin{aligned} \text{logit}(P(Y_{ij} = 1)) = & \beta_0 + \beta_1 I(\text{Baseline is Good}_i) + \beta_2 \text{Treatment}_i \\ & + \beta_3 I(\text{month}_{ij} = 3) + \beta_4 I(\text{month}_{ij} = 4) + \beta_5 \text{Age}(25 \text{ to } 34)_{ij} + \beta_6 \text{Age}(35+)_{ij} \end{aligned}$$

```

gee(formula = nstat ~ baseline + TXT + TIME + as.factor(AGEGROUP),
    id = ID, data = health1, family = "binomial", corstr = "unstructured",
    scale.fix = T, scale.value = 1)

Summary of Residuals:
      Min       1Q   Median       3Q      Max
-0.98144969 -0.18317233  0.08914345  0.17159228  0.83093959

Coefficients:
              Estimate Naive S.E.   Naive z Robust S.E.   Robust z
(Intercept)      0.12457924 0.47137316  0.2642901  0.51374172  0.2424939
baselinePoor     -1.81418056 0.48958528 -3.7055456  0.50961334 -3.5599158
TXTIntervention   2.10225898 0.48779381  4.3097286  0.53777951  3.9091467
TIME              0.03243343 0.03665686  0.8847848  0.04755408  0.6820326
as.factor(AGEGROUP)25-34 1.35250468 0.48130172  2.8100973  0.50420159  2.6824681
as.factor(AGEGROUP)35+  1.42052166 0.79781620  1.7805124  0.78372968  1.8125148

Estimated Scale Parameter: 1
Number of Iterations: 5

Working Correlation
      [,1]      [,2]      [,3]
[1,] 1.0000000 0.1719328 0.5859907
[2,] 0.1719328 1.0000000 0.2013998
[3,] 0.5859907 0.2013998 1.0000000
round(exp(fitb$coefficients),2)
      (Intercept)      baselinePoor      TXTIntervention
      1.13          0.16          8.18
      TIME as.factor(AGEGROUP)25-34 as.factor(AGEGROUP)35+
      1.03          3.87          4.14

```

Based on the output, the interpretations of coefficients estimates are as follows:
The odds ratio of having good self-rating for subjects with good baseline ratings compared with subjects with poor baseline ratings is estimated to be 0.16.
The odds ratio of having good self-rating for subjects in treatment group compared with subjects in control group is estimated to be 8.18.
The odds ratio of having good self-rating is estimated to be 1.03 for each unit of increase in post-randomization month.
The odds ratio of having good self-rating for subjects in 25-34 age group compared with subjects in 15-24 age group is estimated to be 3.87.
The odds ratio of having good self-rating for subjects in 35+ age group compared with subjects in 15-24 age group is estimated to be 4.14.

(c)

$$\begin{aligned}
\text{logit}(P(Y_{ij} = 1|b_i)) &= \beta_0 + \beta_1 I(\text{Baseline is Good}_i) + \beta_2 \text{Treatment}_i \\
&+ \beta_3 I(\text{month}_{ij} = 3) + \beta_4 I(\text{month}_{ij} = 4) + \beta_5 \text{Age}(25 \text{ to } 34)_{ij} + \beta_6 \text{Age}(35+)_{ij} + b_i,
\end{aligned}$$

```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation)
[glmerMod]
Family: binomial ( logit )
Formula: nstat ~ baseline + TXT + TIME + as.factor(AGEGROUP) + (1 | ID)
Data: health1

```

AIC	BIC	logLik	deviance	df.resid
185.0	208.0	-85.5	171.0	192

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6112	-0.2327	0.1402	0.2982	1.8239

Random effects:

Groups Name	Variance	Std.Dev.
ID (Intercept)	5.721	2.392

Number of obs: 199, groups: ID, 78

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.19521	0.87018	0.224	0.82249
baselinePoor	-2.77610	0.98380	-2.822	0.00478 **
TXTIntervention	3.41325	1.07266	3.182	0.00146 **
TIME	0.03718	0.06933	0.536	0.59176
as.factor(AGEGROUP)25-34	2.25651	1.00876	2.237	0.02529 *
as.factor(AGEGROUP)35+	1.98229	1.38118	1.435	0.15123

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.22	2.39	1.25	2.28
baselinePoor	0.06	2.67	0.06	1.00
TXTIntervention	30.36	2.92	24.10	1.00
TIME	1.04	1.07	1.71	1.81
as.factor(AGEGROUP)25-34	9.55	2.74	9.36	1.03
as.factor(AGEGROUP)35+	7.26	3.98	4.20	1.16

Interpretation:

The odds ratio of good self-rating is estimated to be 1.04 for one unit increase in post-randomization month for the same subject.

The main difference between the two interpretations from GEE model and from a linear mixed model is that GEE models make inference about population average while linear mixed model make inference about individuals. The interpretation of linear mixed model should focus on the same subject.